

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

Date: 15 February 2025

Team ID:

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Maximum Marks: 5 Marks

1. Product Backlog, Sprint Schedule, and Estimation

The product backlog below decomposes the HDI Prediction project into epics and user stories, distributed across two sprints, with story points estimated on the Fibonacci scale.

Sprint	Epic	User Story #	User Story / Task	Points	Priority
Sprint-1	Data Collection	USN-1	As a data scientist, I can gather relevant HDI datasets (life expectancy, schooling, GNI per capita) from public sources so that I have raw data to build the model.	2	High
Sprint-1		USN-2	As a data scientist, I can load the collected data into the working environment so that it is ready for preparation.	1	High
Sprint-1	Data Preparation	USN-3	As a data scientist, I can handle missing values in the dataset so that the model is trained on clean, complete data.	3	High
Sprint-1		USN-4	As a data scientist, I can create/engineer new fields from existing indicators so that the model has meaningful features to learn from.	3	Medium
Sprint-1		USN-5	As a data scientist, I can handle inconsistencies in the data so that the dataset is reliable for analysis and modeling.	3	Medium
Sprint-2	Data Visualization	USN-6	As a user, I can view a bar chart of HDI indicators so that I can compare values across countries.	2	Medium
Sprint-2		USN-7	As a user, I can view a pie chart of HDI category distribution so that I can understand the proportion of countries in each tier.	2	Low

Sprint	Epic	User Story #	User Story / Task	Points	Priority
Sprint-2		USN-8	As a user, I can view a line chart of HDI trends so that I can observe changes over time.	2	Medium
Sprint-2		USN-9	As a user, I can view a map visualization of HDI scores so that I can see development levels geographically.	4	Medium
Sprint-2	Dashboard	USN-10	As a user, I can view an interactive dashboard summarizing HDI indicators so that I can explore the data easily.	5	High
Sprint-2	Story	USN-11	As a user, I can enter country indicators (life expectancy, schooling, GNI per capita) and receive a predicted HDI classification (Very High/High/Medium/Low) so that I can assess a country's development level.	5	High
Sprint-3	Model Building	USN-12	As a data scientist, I can train a classification model (e.g., using scikit-learn) on the prepared HDI dataset so that the system can predict a country's HDI category.	5	High
Sprint-3		USN-13	As a data scientist, I can evaluate and tune the model's performance (accuracy, precision, recall) so that predictions are reliable.	3	High
Sprint-3		USN-14	As a data scientist, I can save/export the trained model (e.g., as a pickle file) so that it can be integrated into the web application.	2	Medium
Sprint-3	Application Building	USN-15	As a developer, I can build a Flask web application with input forms for life expectancy, schooling, and GNI per capita so that users can submit country indicators.	3	High
Sprint-3		USN-16	As a developer, I can integrate the trained ML model into the Flask app so that it returns a predicted HDI category when a user submits data.	3	High
Sprint-3	Deployment	USN-17	As a developer, I can deploy the Flask application (e.g., via Anaconda-managed environment) so that users can access the HDI prediction tool.	2	Medium

Sprint Story Point Totals

Sprint	Total Story Points
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Sprint-1	$2 + 1 + 3 + 3 + 3 = 12$
Sprint-2	$2 + 2 + 2 + 4 + 5 + 5 = 20$
Sprint-3	$5 + 3 + 2 + 3 + 3 + 2 = 18$
Total Story Points (Sprint-1 + Sprint-2 + Sprint-3)	$12 + 20 + 18 = 50$

2. Project Tracker, Velocity & Burndown Chart

Project Tracker

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	12	6 Days	01 Jul 2026	06 Jul 2026	12	06 Jul 2026
Sprint-2	20	6 Days	07 Jul 2026	12 Jul 2026		
Sprint-3	18	6 Days	13 Jul 2026	18 Jul 2026		

Velocity

Velocity = Total Story Points Completed / Number of Sprints

Total Story Points = $12 + 20 + 18 = 50$

Number of Sprints = 3

Velocity = $50 / 3 = \text{approx. } 16.7$ Story Points per Sprint

Burndown Chart

A burndown chart is a graphical representation of work remaining versus time. It is commonly used in agile methodologies such as Scrum to track a team's progress toward completing the sprint backlog, and can be applied to this HDI prediction project to track progress across data preparation, visualization, model

building, and deployment.

Recommended references:

- [Visual Paradigm: Scrum Burndown Chart](#)
- [Atlassian: Burndown Charts](#)

References

Project / Technical Resources (HDI Prediction Project)

- [UNDP Human Development Reports — Data Center \(HDI dataset source\)](#)
- [UNDP — Human Development Index methodology](#)
- [scikit-learn documentation](#)
- [pandas documentation](#)
- [NumPy documentation](#)
- [Matplotlib documentation](#)
- [Seaborn documentation](#)
- [Flask documentation](#)
- [Anaconda documentation](#)

Agile / Scrum References

- [Atlassian: Agile Project Management](#)
- [Atlassian: How to do Scrum with Jira Software](#)
- [Atlassian: Epics](#)
- [Atlassian: Sprints](#)
- [Atlassian: Estimation](#)
- [Atlassian: Burndown Charts](#)